Ex. No: 9 Develop a simple calculator using XML-RPC

Aim:

To design a simple calculator using XML-RPC in python based on client-server architecture.

Prerequisites:

Basic understanding of client-server architecture and socket communication.

Knowledge of Python input/output operations, functions definitions, and data types.

Fundamental concept of HTTP and XML.

Algorithm / System Procedure:

Server Program Procedure (calcserver.py)

1. Import simpleXMLRPC server from xmlrpc server
2. Define functions for addition, subtraction, multiplication and division.
3. Check for division by zero in the divide() function.
4. Create the XML-RPC server on localhost and port 8000.
5. Register the calci functions with the server

Client Program Procedure (calcclient.py)

1. Import xmlrpc.client
2. Connect to the server using ServerProxy.
3. Read two floating-point numbers from the user.
4. Invoke the remote calci function.
5. Display the results returned by the server.

Program Code:

Server code: Calcserver.py

```
from xmlrpc.server import SimpleXMLRPCServer
def add(a,b):
    return(a+b)
def subtract(a,b):
    return(a-b)
def multiply(a,b):
    return a*b
def divide(a,b):
    if b==0:
        return "Division by zero not allowed"
    return a/b
server = SimpleXMLRPCServer(("localhost",8000))
print("XML-RPC Calculator Server Running on Port 8000")
server.register_function(add,"add")
server.register_function(subtract,"subtract")
server.register_function(multiply,"multiply")
server.register_function(divide,"divide")
server.serve_forever()
```

Client code: calcclient.py

```
import xmlrpc.client
proxy = xmlrpc.client.ServerProxy("http://localhost:8000/")
a = float(input("Enter First No: "))
b = float(input("Enter Second No: "))
print("\n Addition =", proxy.add(a,b))
print("subtraction =", proxy.subtract(a,b))
print("multiplication =", proxy.multiply(a,b))
print("Division =", proxy.divide(a,b))
```

Execution steps & Sample output:

Step 1. Start the Server Process

python calcserver.py

Server Terminal output:

XML-RPC calculator Server Running on Port 8000

Step 2: Run the Client Process

python calcclient.py

Client Terminal output:

Enter first no: 20

Enter second no: 5

Addition = 25.0

Subtraction = 15.0

Multiplication = 100.0

Division = 4.0

Result:

The XML-RPC based simple calculator application was successfully implemented and executed.